

Javier Gonzalez-Castillo, Ph.D.

Senior Associate Scientist, National Institute of Mental Health, NIH
Washington Metro Area, USA

EDUCATION

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|------|---|
| 2009 | Ph.D., Biomedical Engineering, Purdue University , West Lafayette, IN, USA. |
| 2001 | BS. & MS., Electrical and Computer Engineering, Universidad Politécnica de Madrid , Madrid, Spain. |

RESEARCH EXPERIENCE

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| 2021-Present | <i>Senior Associate Scientist</i>
Section on Functional Imaging Methods, National Institute of Mental Health,
National Institutes of Health , Bethesda, MD. |
| 2024-Present | <i>Affiliated Researcher</i>
Basque Center on Cognition, Brain and Language
San Sebastian, Spain |
| 2025-Present | <i>Volunteer Professor</i>
Department of Biomedical Engineering,
University of Cincinnati , Cincinnati, OH |
| 2014-2021 | <i>Staff Scientist</i>
Section on Functional Imaging Methods, National Institute of Mental Health,
National Institutes of Health , Bethesda, MD. |
| 2011-2014 | <i>Research Fellow</i>
Section on Functional Imaging Methods, National Institute of Mental Health,
National Institutes of Health , Bethesda, MD. |
| 2009-2011 | <i>Post-doctoral Visiting Fellow</i>
Section on Functional Imaging Methods, National Institute of Mental Health,
National Institutes of Health , Bethesda, MD. |

TEACHING EXPERIENCE

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| 2014-2024 | <i>Guest Lecturer at the NIH fMRI Summer Course</i>
National Institute of Mental Health, National Institutes of Health , Bethesda, MD |
| 2008 | <i>Graduate Course Instructor</i>
Weldon School of Biomedical Engineering, Purdue University , West Lafayette <ul style="list-style-type: none">(BME501) Medical Device Accidents and Engineering Analysis |
| 2005-2007 | <i>Teaching Assistant</i>
Weldon School of Biomedical Engineering, Purdue University , West Lafayette <ul style="list-style-type: none">(BME501) Medical Device Accidents and Engineering Analysis(ECE528) Measurement and Stimulation of the Nervous System |

INDUSTRY EXPERIENCE

- 2001-2004 *Associate Consultant*
Hewlett Packard Consulting & Integration, Hewlett-Packard, Madrid, Spain
- Project manager for medium size projects (\$50,000-\$100,000).
 - Prepare commercial proposals on IT management solutions.
 - Manage communication with HP R&D business unit.
- 2000-2001 *Research Engineer*
Hewlett Packard Labs, Hewlett-Packard, Bristol, UK
- Research Semantic Web Technologies and applications to e-commerce.

MENTORSHIP EXPERIENCE

- Post-Doctoral Fellows Somayeh Shahsavarani – NIH & Columbia University (2020 – 2024)
Sharif Kronemer – NIH (2022 – Present)
Fernando Ramirez – NIH (2022 – Present)
Joshua Faskowitz – NIH (2022 – Present)
Tyler Morgan – NIH (2022 – 2024)
- Ph.D. Students Xia Hue – Texas Tech University, Lubbock, TX (2017 – 2018)
Sara Kimmich – NIH/University College of London, London, UK (2017)
Meghan Robinson – Purdue University, West Lafayette, IN (2012)
Samika Kuman – NIH/University of Cambridge, Cambridge, UK (2020-2024)
- NIH Post-bac Fellows Stephanie Swegle (2025-Present), Marly Rubin (2024-Present), Isabel Gephart (2023-2024), Megan Spurney (2021-2023), Isabel Fernandez (2020 – 2022), Ramya Varadarajan (2019 – 2020), Michel Elishama (2018 – 2019), Natasha Topolski (2016 - 2018), Puja Panwar (2016), Laura Buchanan (2014 – 2015), Colin Hoy (2013 - 2014), Kristen Duthie (2011).
- NIH Summer Students James Brown (2016), Devon Shook (2013), Meghan Robinson (2011)

HONORS AND DISTINCTIONS

- 2025 NIMH Special Act Award for providing exceptional mentorship, leadership and stability for staff during a challenging year.
- 2024 NIMH Director's Award in Administrative Excellence: NIHM 75th Anniversary
- 2021 NIMH Outstanding Mentor Award
- 2019 NIMH Director's Award in Administrative Excellence
- 2009 Bilsland Dissertation Fellowship, Purdue University
- 2009 Magoon Award for Excellence in Teaching, Purdue University
- 2008 Magoon Award for Excellence in Teaching, Purdue University
- 2004-2005 **Fulbright Fellowship**

PROFESSIONAL/LEADERSHIP ACTIVITIES

2025-Present	Reviewer for NIH Outstanding Scholars in Neuroscience Awards
2023-Present	Member of the NIMH Office of Fellowship Training Steering Committee
2022-2024	Member of the NIMH 75 th Anniversary Events Working Group
2022	Ad-hoc Reviewer for the National Science Foundation
2016 - 2022	NeuroImage Editorial Board Member
2020 - Present	Editorial Board Member for Frontiers in Brain Imaging Methods
2015 - 2020	Machine Learning-Brain Imaging NIMH's Special Interest Group Organizer Selected Speakers: Dr. Tulay Adali (University of Maryland), Dr. Joshua Vogelstein (John Hopkins University), Dr. Mikail Rubinov (Janelia Farm), Dr. Yoshua Bengio (Montreal University), Dr. Niko Kriegeskorte (Columbia University).
2019, 2023	Ad-hoc Reviewer for the Israel Science Foundation
2020	Topic Editor for Research Topic in Frontiers in Brain Imaging Methods Title: "Paradigm-free functional brain imaging: methods, challenges and opportunities"
2019	Topic Editor for Research Topic in Frontiers in Human Neuroscience Title: "Towards expanded utility of real time fMRI neurofeedback in clinical applications"
2018-2019	Heliyon Editorial Board Member
2018	Symposium Chair: 48 th Meeting for the Society for Neuroscience San Diego, CA Title: "The Dynamic Brain: signatures of fast functional reconfiguration, their interpretability and clinical value"
2017	Local Chair/Organizer for Brainhack Global @ NIMH, Bethesda, MD Two-day Brainhack event held on NIH campus. Projects included: fMRI data denoising, fMRI-based neurofeedback, ME-ICA reporting, AFNI-AROMA integration, and creation of course materials.
2015	Grant Review Panelist National Science Foundation (NSF) & National Institutes of Health (NIH) <u>Panel:</u> Collaborative Research in Computational Neuroscience
2014	Grant Review Panelist Army Research Lab (ARL) <u>Panel:</u> Cognition and Neuroergonomics Collaborative Technology Alliance.
2012-2020	Abstract Reviewer Annual Meeting of the International Society for Magnetic Resonance in Medicine (ISMRM)
2009-2020	Abstract Reviewer Annual Meeting of the Organization for Human Brain Mapping (OHBM)
2009-Present	Scientific Journal Reviewer PNAS, Journal of Neuroscience, Trends in Cognitive Neuroscience, Nature, Nature Neuroscience, Nature Communications, NeuroImage, NeuroImage: Clinical,

Human Brain Mapping, Neuroscience and Behavioral Reviews, Network Neuroscience, Magnetic Resonance in Medicine, IEEE Transactions in Biomedical Engineering, Frontiers in Neuroscience, Frontiers in Brain Imaging Methods, Artificial Intelligence in Medicine, Neuroscience Letters, Cognitive Neuroscience, Neuroscience Research, SPIE Journal of Medical Imaging, Brain Structure and Function, Neuroinformatics.

OUTREACH ACTIVITIES

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| 2022-2024 | Organizer of tour of NIMH Functional Neuroimage Facilities for recipients of the OSNAP Award |
| 2019-Present | Mentor at the National Research Mentoring Network
Mentees: Dr. Serenella Tolomeo (2019) |

PROFESSIONAL MEMBERSHIPS

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| 2012-Present | <i>Society for Neuroscience (sfn)</i> |
| 2011-2019 | <i>International Society for Resonance Magnetic Imaging in Medicine (ISMRM)</i> |
| 2011-2022 | <i>Organization for Human Brain Mapping (OHBM)</i> |

HONORARY SOCIETY MEMBERSHIPS

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| 2008-Present | <i>Tau Beta Pi, Engineering Honor Society</i> |
| 2008-Present | <i>Golden Key, International Honor Society</i> |

GRANTS

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| 2017 - 2020 | NIMH Scientific Director's call for Scientific Workshops and Talk Series
"Talk Series on Machine Learning in Brain Imaging and Neuroscience"
Support: \$20,000 / year |
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BOOK CHAPTERS

- [1] **Gonzalez-Castillo J**, Bandettini PA. "*Task-based Dynamics: a systems-level functional connectivity perspective*" in Calhoun V. eds. *The Chronnectome*. Elsevier (Under review)
- [2] **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA. "*How to design and implement Resting-state functional Magnetic Resonance Imaging studies*" in Chrysikou E. eds. *Cognitive Neuroscience Methods*. Springer Nature (Accepted pending minor revisions)
- [3] Malaia E, **Gonzalez-Castillo J**, Webber-Fox C, Talavage TM, Wilbur B. "*Neuronal processing of verbal event structure: temporal and functional dissociation between telic and atelic verbs*". In: de Almeida RG, Manouilidou C, eds. *Cognitive science perspectives on verb representation and processing*. Cham: Springer International Publishing; **2015**:131–140.
- [4] Talavage TM, Johnsrude IS, **Gonzalez-Castillo J**. "*Hemodynamic Imaging: Functional Magnetic Resonance Imaging*". In: Poeppel D, Overath T, Popper AN, Fay RR, eds. *The Human Auditory Cortex*. Vol 43. Springer Handbook of Auditory Research. New York, NY: Springer New York; **2012**:129–162.

SELECTED RESEARCH PUBLICATIONS

- [1] **Gonzalez-Castillo J**, Spurney M, Lam KC, Pereira F, Handwerker DA, Bandettini PA. “*In-scanner thoughts contribute to resting-state functional connectivity*” *Nature Communications*, **2026** (in Press)
- [2] Kronemer SI, Bandettini PA, **Gonzalez-Castillo J**. “*Sleuthing subjectivity: emerging covert measures of consciousness*” *Nature Reviews Neuroscience*, **2025**.
- [3] **Gonzalez-Castillo J**. “*Traveling and Standing waves in the brain*” *Nature Neuroscience*, **2022**;25(8), 980-981.
- [4] **Gonzalez-Castillo J**, Kam JWY, Hoy CW, Bandettini PA. “*How to interpret resting-state fMRI: ask your participants*” *Journal of Neuroscience*, **2020**;41(6), 1130-1141
- [5] **Gonzalez-Castillo J**, Caballero-Gaudes C, Topolski N, Handwerker DA, Pereira F, Bandettini PA. “*Imaging the spontaneous flow of thought: distinct periods of cognition contribute to dynamic functional connectivity during rest*”. *NeuroImage*, **2019**; 202, 116129
- [6] **Gonzalez-Castillo J**, Bandettini PA. “*Task-based dynamic connectivity: recent findings and open questions*” *NeuroImage*, **2018**; 180, 526-533
- [7] Saggar M, Sporns O, **Gonzalez-Castillo J**, Bandettini PA, Carlsson G, Glover G, Reiss AL. “*Towards a new approach to reveal dynamical organization of the brain using topological data analysis*”. *Nature Communications*, **2018**; 9(1):1399
- [8] **Gonzalez-Castillo J**, Hoy CW, Handwerker D, Robinson ME, Buchanan LC, Saad ZS, Bandettini PA. “*Tracking ongoing in individuals using brief, whole-brain functional connectivity patterns*” *Proc Natl Acad Sci U S A*. **2015**; 112(28): 8762-8767
- [9] **Gonzalez-Castillo J**, Bandettini PA. “*What cascade spreading models can tell us about the brain*” *Neuron*. **2015**; 86(6): 1327-1329
- [10] **Gonzalez-Castillo J**, Saad ZS, Handwerker DA, Inati SJ, Brenowitz N, Bandettini PA. “*Whole-brain, time-locked activation with simple tasks revealed using massive averaging and model-free analysis*” *Proc. Natl. Acad. Sci. USA* **2012**; 109(14): 5487–92

ADDITIONAL RESEARCH PUBLICATIONS

- [11] Kumar S, Arzi A, Bareham CA, **Gonzalez-Castillo J**, et al. “*Brain Network Dynamics in the Wake-Sleep Transition according to task engagement*” *Imaging Neuroscience*, **2026** (In Press)
- [12] Lyu R, Bari S, Goñi J, **Gonzalez-Castillo J**, Talavage TM “*Reference Centroid Selection Contributes to Self-Identification and Harmonization of Tangent-Space Functional Connectomes*” *Network Neuroscience* (Under review)
- [13] **Gonzalez-Castillo J**, Caballero-Gaudes C, Handwerker DA, Bandettini PA. “*A new fMRI quality metric using multi-echo information: Theory, validation and implications*” *Imaging Neuroscience* (under review) / *bioRxiv*, **2026**.

- [14] Uruñuela E, **Gonzalez-Castillo J**, Aulakh A, Handwerker DA, Wilms M, Forkert ND “A multimodal spatio-temporal CNN for ICA component classification in multi-echo fMRI” *Network Neuroscience* (Under review).
- [15] Miyawaki Y, Koise K, Handwerker DA, **Gonzalez-Castillo J** et al. “Rapid decoding of neural information representation from ultra-fast functional magnetic resonance imaging signals” (Under review)
- [16] Faskowitz J, Bandettini PA, **Gonzalez-Castillo J**. “Mapping high-amplitude fMRI edge time series events across space and time” *Imaging Neuroscience*, **2026** 4:IMAG.a.1126.
- [17] Taylor PA, Aggarwal H et al. “Go Figure: Transparency in neuroscience images preserves context and clarifies interpretation” *Nature Methods* (In Press).
- [18] Kronemer SI, Gobo VE, Teves JB et al. “Cross-species real time detection of trends in pupil size fluctuations” *Behavior Research Methods*, **2025**: 57(1), 1 -14.
- [19] Kronemer SI, Holness M, Morgan AT et al. “Visual imagery vividness correlates with afterimage brightness and sharpness” *Neuroscience of Consciousness*, **2024**:1,niae032
- [20] Gaggi NL, Collins KA, **Gonzalez-Castillo J**, Hurtado AM et al. “Transcranial photobiomodulation increases brain activity within irradiated areas in early Alzheimer’s disease: Potential link to cerebral metabolism” *Brain Stimulation*, **2024**:17,208-210.
- [21] Revsine C, **Gonzalez-Castillo J**, Merriam EP, Bandettini PA, Ramirez FM. “A unifying model for discordant and concordant results in human neuroimaging studies of facial viewpoint selectivity” *J Neuroscience*, **2024**:e029623202
- [22] Uruñuela E, **Gonzalez-Castillo J**, Zheng C, Bandettini PA, Caballero-Gaudes C. “Whole-brain multivariate hemodynamic deconvolution for functional MRI with stability selection” *IEEE Transactions in Medical Imaging*, **2024**:91, 103010
- [23] **Gonzalez-Castillo J**, Fernandez IS, Lam KC, Handwerker DA, Pereira F, Bandettini PA “Manifold learning for fMRI time-varying FC” *Front. Human Neuros.*, **2023**:17,1-22
- [24] Shahsavarani S, Thibodeaux DN, Xu W, et al. “Cortex-wise neural dynamics predict behavioral states and provide a neural basis for resting-state dynamic functional connectivity” *Cell Reports*, **2023**:42(6), 112527
- [25] Taylor PA, Reynolds RC, Calhoun V, **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA, Mejia AF, Chen G. “Highlight results, Don’t hide them: Enhance interpretation, reduce biases and improve reproducibility” *NeuroImage*, **2023**:274, 120138
- [26] Teves JB, **Gonzalez-Castillo J**, Holness M, Spurney M, Bandettini PA, Handwerker DA. “The art and science of using quality control to understand and improve fMRI data”. *Front. Neurosci.*, **2023**:17, 1100544
- [27] Faskowitz J, Moyer D, Handwerker DA, **Gonzalez-Castillo J** et al. “Commentary on Pang et al. (2023) *Nature*” *BioRxiv*, **2023**.07.20.549785
- [28] Bandettini PA, **Gonzalez-Castillo J**, Handwerker DA, Taylor P, Chen G, Thomas A. “The challenge of BWAS: Unknown unknowns in feature space and variance” *Med*, **2022**: 3 (8), 526-531.
- [29] **Gonzalez-Castillo J**, Fernandez I, Handwerker DA, Bandettini PA. “Ultra-slow fMRI fluctuations in the fourth ventricle as a marker of drowsiness” *NeuroImage*, **2022**: 259, 119424

- [30] DuPre E, Salo T, Ahmed Z, Bandettini PA et al. “*TE-dependent analysis of multi-echo fMRI with tedana*” *Journal of Open Source Software*, **2021**; 6 (66), 3669.
- [31] Gau R, Noble S, Heuer K, et al. “*Brainhack: developing a culture of open, inclusive, community-driven neuroscience*” *Neuron*, **2021**;109(11),1769-1775
- [32] **Gonzalez-Castillo J**, Ramot M, Momenan R. “*Towards Expanded Utility of Real Time fMRI Neurofeedback in Clinical Applications*” *Frontiers in Human Neuroscience*, **2020**; 14, 471
- [33] Rolinski R, You X, **Gonzalez-Castillo J**, Norato G, Reynolds RC, Inati SK, Theodore WH. “*Language lateralization from task-based and resting-state functional MRI in patients with epilepsy*”. *Human Brain Mapping*, **2020**; 41(11), 3133-3146
- [34] Handwerker DA, Ianni G, Gutierrez B, Roopchansingh V, **Gonzalez-Castillo J**, Chen G, Bandettini PA, Ungerleider LG, Pitcher D. “*Theta-burst TMS to the posterior superior temporal sulcus decreases resting-state fMRI connectivity across the face processing network*”. *Network Neuroscience*, **2020**; 4(3), 746-760
- [35] Chai Y, Handwerker DA, Marrett S, **Gonzalez-Castillo J**, Merriam EP, Hall A, Molfese PJ, Bandettini PA. “*Visual temporal frequency shows a distinct cortical architecture using fMRI*” *NeuroImage*, **2019**; 197, 13-23
- [36] Caballero-Gaudes C, Moia S, Panwar PA, Bandettini PA, **Gonzalez-Castillo J**. “*A deconvolution algorithm for multi-echo functional MRI: multi-echo sparse paradigm free mapping*” *NeuroImage*, **2019**; 202, 116081
- [37] Xie H, Zheng CY, Handwerker DA, Bandettini PA, Calhoun VD, Mitra S, **Gonzalez-Castillo J**. “*Efficacy of different dynamic functional connectivity methods to capture cognitively relevant information*”. *NeuroImage*, **2019**; 188, 502-514
- [38] Ramot M, **Gonzalez-Castillo J**. “*A framework for offline evaluation and optimization of real-time algorithms for use in neurofeedback, demonstrated on an instantaneous proxy for correlations*” *NeuroImage*, **2019**; 188, 322-334
- [39] Xie H, **Gonzalez-Castillo J**, Damaraju E, Bandettini PA, Calhoun V, Mitra S. “*Time-varying whole-brain functional network connectivity coupled to task engagement*” *Network Neuroscience* **2018**; 3(1):49-66
- [40] Xie H, Calhoun V, **Gonzalez-Castillo J**, Damaraju E, Miller R, Bandettini PA, Mitra S. “*Whole-brain connectivity dynamics reflect both task-specific and individual-specific modulation: a multitask study*”. *NeuroImage*, **2018**; 180 (Part B), 495-504
- [41] Jangraw DC, **Gonzalez-Castillo J**, Handwerker DA, Ghane M, Rosenberg M, Panwar P, Bandettini PA. “*A Functional Connectivity-based neuromarker of Sustained Attention Generalizes to Predict Recall in Naturalistic Reading Task*” *NeuroImage* **2018**; 166:99-109
- [42] Torrisi S, Gorka AX, **Gonzalez-Castillo J**, O’Connell K, Balderston N, Grillon C, Ernst M. “*Extended amygdala connectivity changes during sustained shock anticipation*” *Translational Psychiatry* **2018**; 8(1):33
- [43] Huber L, Handwerker D, Jangraw D, Hall H, Stuber C, **Gonzalez-Castillo J**, Ivanov D, Marrett S, Guidi M, Goense J, Poser BA, Bandettini PA. “*High-Resolution CBV-fMRI allows mapping of laminar activity and connectivity of cortical input and output in human M1*” *Neuron* **2017**; 96(6):1253-1263.e7

- [44] Ramot M, Kimmich S, **Gonzalez-Castillo J**, Roopchansingh V, Popal H, White E, Gotts S, Martin A. “Direct modulation of aberrant brain network connectivity through real-time neurofeedback” *eLife* **2017**; 6:e28974
- [45] **Gonzalez-Castillo J**, Gang C, Nichols T, Cox B, Bandettini PA. “Variance Decomposition for multi-session fMRI”. *NeuroImage*, Special Issue on “Cleaning up the fMRI timeseries” *NeuroImage* **2017**; 154: 206-218
- [46] Degryse J, Seurinck R, Durnez J, **Gonzalez-Castillo J**, Bandettini PA, Moerkerke B. “Introducing alternative-based thresholding for defining functional regions of interest in fMRI” *Front. Neurosci.* **2017**; 11:222
- [47] Craddock RC, Bellec P, Margules DS, Nichols BN, Pfannmöller JP, Badhwar AP, Kennedy D, Poline JB, Toro R, Cipollini B, Rokem A et al. “2015 Brainhack Proceedings” *GigaScience* **2016**; 5(1): 1 – 26
- [48] **Gonzalez-Castillo J**, Caballero Guades C, Panwar P, Buchanan LC, Handwerker DA, Jangraw DC, Zachariou V, Bandettini PA. “Evaluation of Multi-Echo ICA denoising for task based fMRI studies: block designs, rapid event-related designs, and cardiac-gated fMRI” *NeuroImage* **2016**; 141: 452-468.
- [49] **Gonzalez-Castillo J**, Hoy CW, Handwerker DA, et al. “Task Dependence, Tissue Specificity, and Spatial Distribution of Widespread Activations in Large Single-Subject Functional MRI Datasets at 7T” *Cereb Cortex.* **2015**; 25(12): 4667-4677
- [50] Yang Z, Huang Z, **Gonzalez-Castillo J**, Dai R, Northoff G, Bandettini PA. “Using fMRI to decode true thoughts independent of intention to conceal” *NeuroImage.* **2014**; 99: 80–92.
- [51] **Gonzalez-Castillo J**, Handwerker DA, Robinson ME, et al. “The spatial structure of resting state connectivity stability on the scale of minutes” *Front Neurosci.* **2014**; 8(June): 138
- [52] Talavage TM, **Gonzalez-Castillo J**, Scott SK. “Auditory neuroimaging with fMRI and PET” *Hear Res.* **2014**; 307: 4–15.
- [53] Smalt CJ, **Gonzalez-Castillo J**, Talavage TM, Pisoni DB, Svirsky MA. “Neural correlates of adaptation in normal hearing subjects to free learning with cochlear implant acoustic simulations” *NeuroImage* **2013**; 82: 500-509
- [54] Hutchison RM, Womelsdorf T, Allen EA, Bandettini PA, Calhoun VD, Corbetta M, Della Penna, S, Dyun, JH, Glover GH, **Gonzalez-Castillo J**, et al. “Dynamic functional connectivity: Promise, issues, and interpretations” *NeuroImage* **2013**; 80: 360–378.
- [55] Bandettini PA, Kundu P, **Gonzalez-Castillo J**, Misaki M, Guillod P. “Characterizing and utilizing fMRI fluctuations, patterns, and dynamics” In: Weaver JB, Molthen RC, eds. *SPIE Medical Imaging*. International Society for Optics and Photonics; **2013**: 86720T
- [56] **Gonzalez-Castillo J**, Duthie KN, Saad ZS, Chu C, Bandettini PA, Luh W-M. “Effects of image contrast on functional MRI image registration” *NeuroImage* **2012**; 67: 163–174.
- [57] Handwerker DA, Roopchansingh V, **Gonzalez-Castillo J**, Bandettini PA. “Periodic changes in fMRI connectivity”. *NeuroImage* **2012**; 63(3): 1712–9.
- [58] Handwerker DA, **Gonzalez-Castillo J**, D’Esposito M, Bandettini PA. “The continuing challenge of understanding and modeling hemodynamic variation in fMRI” *NeuroImage.* **2012**; 62(2): 1017–1023.

- [59] **Gonzalez-Castillo J**, Olulade OA, Talavage TM. “Using functional MRI to study auditory comprehension” *Imaging Med.* **2012**; 4(1): 137–143
- [60] Olulade O, Hu S, **Gonzalez-Castillo J**, et al. “Assessment of temporal state-dependent interactions between auditory fMRI responses to desired and undesired acoustic sources” *Hear. Res.* **2011**; 277(1-2): 67–77.
- [61] Soltysik DA, Thomasson D, Rajan S, **Gonzalez-Castillo J**, DiCamillo P, Biassou N. “Head-repositioning does not reduce the reproducibility of fMRI activation in a block-design motor task” *NeuroImage* **2011**; 56(3): 1329–37.
- [62] **Gonzalez-Castillo J**, Roopchansingh V, Bandettini PA, Bodurka J. “Physiological noise effects on the flip angle selection in BOLD fMRI” *NeuroImage* **2011**; 54(4): 2764–2778.
- [63] **Gonzalez-Castillo J**, Talavage TM. “Reproducibility of fMRI activations associated with auditory sentence comprehension” *NeuroImage* **2011**; 54(3): 2138–2155.
- [64] Hu S, Olulade O, **Gonzalez-Castillo J**, et al. “Modeling hemodynamic responses in auditory cortex at 1.5T using variable duration imaging acoustic noise” *NeuroImage* **2010**; 49(4): 3027–3038.
- [65] Kemmerer D, **Gonzalez-Castillo J**. “The Two-Level Theory of verb meaning: An approach to integrating the semantics of action with the mirror neuron system” *Brain Lang.* **2010**; 112(1): 54–76.
- [66] Kemmerer D, **Gonzalez-Castillo J**, Talavage T, Patterson S, Wiley C. “Neuroanatomical distribution of five semantic components of verbs: Evidence from fMRI” *Brain Lang.* **2008**; 107(1): 16–43.
- [67] **Gonzalez-Castillo J**, Trastour D, Bartolini C. “Description Logics for Matchmaking of Services” In: KI-2001 Workshop on Applications of Description Logics, Vienna, Austria. **2001**
- [68] Trastour D, Bartolini C, **Gonzalez-Castillo J**. “Semantic Web Approach to Service Description for Matchmaking of Services” In: *Semantic Web Workshop*, Stanford USA. **2001**

CONFERENCE ARTICLES

- [1] Caballero-Gaudes C, Moia S, Bandettini, PA, **Gonzalez-Castillo J**. “Quantitative deconvolution of fMRI data with multi-echo sparse paradigm free mapping” 21st International Conference on Medical Imaging Computing and Computer Assisted Intervention (MICCAI), Granada, Spain, **2018**, 311-319
- [2] Caballero Guades C, Bandettini PA, **Gonzalez-Castillo J**. “A temporal deconvolution algorithm for multiecho functional MRI” IEEE International Symposium on Biomedical Engineering (ISBI), Washington DC, **2018**, 608-611.

ORAL PRESENTATIONS

- [1] “Resting-state fMRI” NIMH fMRI Summer course, Bethesda, MD, July, **2026**.
- [2] “Realtime fMRI and Neurofeedback” NIMH fMRI Summer course, Bethesda, MD, July, **2026**.
- [3] “Role of In-scanner Conscious Experience in Resting-state” Annual Meeting of the Association for the Scientific Study of Consciousness, Santiago de Chile, Chile, July, **2026**.

- [4] “Now that I have Multi-Echo data, what else can I do with it?” Annual Meeting of the Organization for Human Brain Mapping, Bordeaux, France, June, **2026**.
- [5] “Role of spontaneous conscious process in resting-state fMRI” Biomedical Engineering Talk Series, University of Cincinnati, Cincinnati, OH, April **2026**.
- [6] “Inner-experience during rest: implications for fMRI research” *Perception in the Primate Brain Workshop*, NIH Research Festival, NIH, Bethesda, MD, **2025**.
- [7] Keynote: “Multi-echo fMRI”, *2024 High Performance Neuroimaging Symposium*, Madison, WI, September, **2024**.
- [8] “Connectivity and Dynamic Connectivity” *NIMH Summer fMRI Course*, NIH, Bethesda, MD, September, **2024**.
- [9] “Resting-state fMRI: brief overview of methods and interpretational challenges” *NIH Pain Special Interest Group Seminar Series*, Bethesda, MD, April, **2024**.
- [10] “Contribution of In-Scanner Thoughts to Resting-State Functional Connectivity: how participants rest matters”. *Annual Meeting of the Society for Neuroscience*, Washington, DC, November, **2023**.
- [11] “Participant introspection to understand resting-state fMRI” ISMRM Workshop on Current Issues in Brain Function”, Padua, Italy, September, **2023**.
- [12] “How subjects rest in the scanner matters”, Basque Center on Cognition, Brain and Language, San Sebastian, Spain, May, **2023**.
- [13] “How subjects rest in the scanner matters”, CAMRI Neuroimaging Seminar Series, Baylor College of Medicine, Houston, Texas, USA, April, **2023**.
- [14] “Ultra-slow fMRI fluctuations in the fourth ventricle as a marker of drowsiness” Laboratory of Neuroimaging, NIAAA, NIH, Bethesda, October, **2022**.
- [15] “What to do when stimuli are missing: advanced methods for time-varying resting-state fMRI” Neuroscape Center, University of California, San Francisco, CA. July **2022**.
- [16] “Sources of individual differences in resting-state fMRI” *NIMH Workshop on Neuroimaging of Individual Differences and Naturalistic Stimuli*, virtual, August, **2021**
- [17] “Cognitive correlates of BOLD Resting-state dynamic functional connectivity” *Scientific Workshop: Brain Functional Organization, Connectivity and Behavior*, Whistler, CA, March, **2020**
- [18] “Periods of discernible cognition contribute to dynamic functional connectivity during rest” *27th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Montreal, May, **2019**.
- [19] “Quantitative deconvolution of neuronal-related BOLD events with multi-echo sparse free paradigm mapping” *26th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Paris, June, **2018**.
- [20] “Efficacy of different functional connectivity methods to capture cognitively relevant information” *26th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Paris, June, **2018**.

- [21] “Machine Learning in NeuroImaging” *NIH fMRI Summer Course, National Institutes of Health, Bethesda, MD. August, 2017.*
- [22] “Unconventional fMRI Methodology: multi-echo fMRI, connectivity dynamics, and fMRI-neurofeedback” *Centro Integral de Neurociencias, Hospital de Madrid, Madrid, Spain. December, 2016.*
- [23] “Alternative analyses for task-based fMRI” *24th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine, Singapore. May, 2016.*
- [24] “Wide-spread brain activation and connectivity dynamics with BOLD fMRI” *Indiana University, Bloomington, IN. April, 2016.*
- [25] “Wide-spread brain activation and connectivity dynamics with BOLD fMRI” *Texas Tech University, Lubbock, TX. March, 2016.*
- [26] Keynote: “Multi-echo EPI for resting state and activation-based fMRI” *Texas Tech NeuroImaging Institute, Lubbock, TX. March, 2016.*
- [27] “The richness of the BOLD signal: Challenges and Opportunities” *Cincinnati Children’s Hospital, Cincinnati, OH. March, 2016.*
- [28] “Wide-spread brain activation and functional connectivity decoding with BOLD fMRI” *National Library of Medicine, Bethesda, MD. February, 2016.*
- [29] “Multi-echo EPI for resting state and activation based fMRI” *NIH fMRI Summer Course, National Institutes of Health, Bethesda, MD. August, 2015.*
- [30] “Dynamic Resting State fMRI Assessment” *NIH fMRI Summer Course, National Institutes of Health, Bethesda, MD. August, 2015.*
- [31] “fMRI-based functional connectivity: Issues and Applications” *Georgetown University, Washington, DC. August, 2015*
- [32] “Wide-spread brain activation with BOLD fMRI” *Max Planck Institute, Leipzig, Germany. July, 2015.*
- [33] “Optimizing fMRI data acquisition and analysis” *Basque Center for Brain and Cognition, San Sebastian, Spain. July, 2015.*
- [34] “Realtime fMRI and Neurofeedback” *NIH fMRI Summer Course, National Institutes of Health, Bethesda, MD. August, 2014.*
- [35] “BOLD resting state dynamics and its relationship to on-going cognition” *1st International Conference on Brain Development, Beijing, China. August, 2014.*
- [36] “Resting State Connectivity Dynamics: Basic Characterization and Relationship to Cognition” *National Institutes of Health, Bethesda, MD. August 2013.*
- [37] “Understanding Resting State fMRI Connectivity Dynamics” *National Institute of Drug Abuse, Baltimore, MD. July 2013.*
- [38] “When does a task disturb rest?” *19th Annual Meeting of the Organization for Human Brain Mapping, Seattle, WA. June 2013.*

- [39] “Optimization of acquisition and analysis procedures for BOLD temporal series” *Fundación Centro Investigación Enfermedades Neurológicas (CIEN)*, Reina Sofia Alzheimer’s Research Center, Madrid, Spain. January **2013**.
- [40] “Is the sparseness of fMRI activation maps real or a result of insufficient TSNR?” *Institute of Psychology, Chinese Academy of Sciences*, Beijing, China. June **2012**.
- [41] “What is the ultimate sensitivity of fMRI: Does the whole brain activate?” *20th Annual Meeting for the International Society of Magnetic Resonance in Medicine*. Melbourne, Australia. May **2012**.
- [42] “fMRI reveals whole-brain time-locked activations to simple tasks using high-order averaging and model-free analysis” *John Hopkins University*, Baltimore, MD. April **2012**.
- [43] “Dealing with physiological noise, TSNR, and how to easily improve alignment of fMRI and anatomical data” *West Virginia University*, Morgantown, WV. April **2012**.
- [44] “Realtime fMRI and fMRI Neurofeedback” *West Virginia University*, Morgantown, WV. April **2012**.
- [45] “What is the ultimate sensitivity of fMRI: Does the whole brain activate?” *National Institute of Health, fMRI/MRI Series*, Bethesda, MD, **2011**.
- [46] “BOLD responses to a simple visual stimulation + attention control task can be detected in over 90% of the brain when TSNR is sufficiently high” *15th Annual NIMH/DIRP Scientific Retreat*, Lancaster, PA, **2011**.
- [47] “Longitudinal fMRI Study of Adaptation to Degraded Speech Stimuli” *1st Indiana Neuroimaging Symposium*, Bloomington, IN, **2007**.
- [48] “Event related fMRI of Adaptation to Acoustic Simulation of Cochlear Implant Electrical Stimulation” *29th Midwinter Meeting of the Association for Research in Otolaryngology*, Baltimore, MD, **2006**.

POSTER PRESENTATIONS

- [1] Levesque CM, Kronemer SI, **Gonzalez-Castillo J** et al. “A multimodal neuroimaging case study of blindsense in cortical blindness” *32th Annual Meeting of the Organization for Human Brain Mapping*, Bordeaux, France, June **2026**.
- [2] Uruñuela E, **Gonzalez-Castillo J**, Aulakh A et al. “Separation of Mixed Components in Multi-Echo fMRI ICA Using Conditional Diffusion Models” *32th Annual Meeting of the Organization for Human Brain Mapping*, Bordeaux, France, June **2026**.
- [3] Faskowitz J, PA Bandettini, **Gonzalez-Castillo J** “Spatial maps of similarity across the cortex” *32th Annual Meeting of the Organization for Human Brain Mapping*, Bordeaux, France, June **2026**.
- [4] **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA “Evaluation of BOLD contributions to the Global Signal and fluid compartments with multi-echo fMRI” *32th Annual Meeting of the Organization for Human Brain Mapping*, Bordeaux, France, June **2026**.
- [5] Tibblin SF, **Gonzalez-Castillo J** et al. “Does cerebrovascular reactivity lag heterogeneity underlie breath-hold RSN mimicry?” *2026 Annual Meeting of the International Society for Magnetic Resonance Imaging in Medicine*, Cape Town, South Africa, May **2026**

- [6] **Gonzalez-Castillo J**, Caballero-Gaudes C, Handwerker DA, Bandettini PA. “Utility of across-echo connectivity as a Quality Control Metric for fMRI” *54th Annual Meeting of the Society for Neuroscience*, San Diego, USA, November **2025**.
- [7] Kronemer SI, Wang A, Walsh C et al. “Covertly predicting and augmenting conscious perception in cerebral blindness” *54th Annual Meeting of the Society for Neuroscience*, San Diego, USA, November **2025**.
- [8] Suwanamalik-Murphy P, Kronemer SI, Gobo VE, **Gonzalez-Castillo J** et al. “Investigating eye metrics as an indicator of sensory neural processing, behavior, and conscious perception” *54th Annual Meeting of the Society for Neuroscience*, San Diego, USA, November **2025**.
- [9] Faskowitz J, Bandettini PA, **Gonzalez-Castillo J** “Modeling functional connectivity as brief events” *54th Annual Meeting of the Society for Neuroscience*, San Diego, USA, November **2025**.
- [10] R. Lyu, T. Talavage and **J. Gonzalez-Castillo**, "Tangent-Space Analysis of Resting-State Functional MRI Connectomes for Detection of Abnormal Brain Health in High School Collision-Sport Athletes," Biomedical Engineering Society 2025 Annual Meeting, San Diego, USA, October, **2025**.
- [11] Faskowitz J, **Gonzalez-Castillo J**, Bandettini PA. “Edge event organization across temporal categories” *31th Annual Meeting of the Organization for Human Brain Mapping*, Brisbane, Australia, June **2025**.
- [12] **Gonzalez-Castillo J**, Caballero-Gaudes C, Handwerker DA, Bandettini PA. “Multi-echo Functional Connectivity as an evaluation metric for denoising” *31th Annual Meeting of the Organization for Human Brain Mapping*, Brisbane, Australia, June **2025**.
- [13] Handwerker DA, Uruñuela E, Abbott D et al. “Tedana: expanding flexibility for multi-echo fMRI noise removal methods” *31th Annual Meeting of the Organization for Human Brain Mapping*, Brisbane, Australia, June **2025**.
- [14] Tibblin SF, **Gonzalez-Castillo J**, Hocke LM, Miller RL et al. “Precision functional mapping of cerebrovascular reactivity networks” ISMRM Joint Workshop of the Ultra-high field MR and Brain function study groups, Annapolis, USA, April **2025**.
- [15] Gephart IS, Gonzalez-Castillo J, Spurney MA et al. “Removal of slow, brain-wide spatiotemporal patterns improves predictions of in-scanner experience during rest” *17th Annual Meeting of the Social and Affective Neuroscience Society*, Chicago, April **2025**.
- [16] **Gonzalez-Castillo J**, Gephart I, Spurney M, Handwerker DA, Bandettini PA “Contribution of slow, brain-wide waves of activity to spontaneous thoughts during resting-state” *53rd Annual Meeting of the Society for Neuroscience*, Chicago, USA, October **2024**.
- [17] Handwerker DA, Bandettini PA, Dowdle L, DuPre E, **Gonzalez-Castillo J** et al. “*tedana: Multi-echo fMRI noise removal software and resources*” *30th Annual Meeting of the Organization for Human Brain Mapping*, Seoul, South Korea, July **2024**.
- [18] Faskowitz J, **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA “On the features of spiking connectivity” *30th Annual Meeting of the Organization for Human Brain Mapping*, Seoul, South Korea, July **2024**.

- [19] Gobo V, **Gonzalez-Castillo J**, Teve J, Holness M, Bandettini PA, Kronemer S “Phase of pupillary unrest corresponds with perceptual sensitivity, MEG and whole brain fMRI signals” *30th Annual Meeting of the Organization for Human Brain Mapping*, Seoul, South Korea, July **2024**.
- [20] Kronemer S, Holness M, Akin B, **Gonzalez-Castillo J**, et al. “Whole brain and primary visual cortex layer fMRI signatures of afterimage perception” *30th Annual Meeting of the Organization for Human Brain Mapping*, Seoul, South Korea, July **2024**.
- [21] Gephart I, **Gonzalez-Castillo J**, Spurney M, Handwerker DA, Bandettini PA “Contributions of slow, brain-wide patterns of activity to ongoing experience in resting-state fMRI” *30th Annual Meeting of the Organization for Human Brain Mapping*, Seoul, South Korea, July **2024**.
- [22] Morgan AT, Gephart I, Handwerker DA, **Gonzalez-Castillo J**, Bandettini PA “A functionally time-resolved reconstruction technique for high-resolution fMRI (fTR_MRI)” *52nd Annual Meeting of the Society for Neuroscience*, Washington, DC, USA, November **2023**.
- [23] **Gonzalez-Castillo J**, Spurney M, Lam KC, Pereira F, Handwerker DA, Bandettini PA. “Contribution of In-scanner Thoughts to Resting-state Functional Connectivity: How participants rest matters” *52nd Annual Meeting of the Society for Neuroscience*, Washington, DC, USA, November **2023**.
- [24] Spurney M, Faskowitz J, Gonzalez-Castillo J, Handwerker DA, **Bandettini PA**. “Evaluating the predictive power of dynamic fMRI connectivity summary metrics” *52nd Annual Meeting of the Society for Neuroscience*, Washington, DC, USA, November **2023**.
- [25] Gephart I, Morgan TA, **Gonzalez-Castillo J**, Bandettini PA. “Investigating the neural correlates of the audiovisual bounce effect using fMRI” *52nd Annual Meeting of the Society for Neuroscience*, Washington, DC, USA, November **2023**.
- [26] Kronemer SI, Holness M, Morgan TA, **Gonzalez-Castillo J**, Teves JB, Akin B, Huber R, Gobo VE, Handwerker DA, Bandettini PA. “Perceptually-matched images and afterimages share whole brain fMRI dynamics” *52nd Annual Meeting of the Society for Neuroscience*, Washington, DC, USA, November **2023**.
- [27] Gobo VE, **Gonzalez-Castillo J**, Teves JB, Holness M, Bandettini PA, Kronemer SI “Pupil size and phase as a real-time marker of perceptual sensitivity and whole-brain activity” *52nd Annual Meeting of the Society for Neuroscience*, Washington, DC, USA, November **2023**.
- [28] Ramirez FM, **Gonzalez-Castillo J**, Bandettini PA “Empirical evidence supports low-level account of observations of mirror-symmetry in the representation of facial viewpoint in human-selective areas” *52nd Annual Meeting of the Society for Neuroscience*, Washington, DC, USA, November **2023**.
- [29] Fan DJ, **Gonzalez-Castillo J**, Bandettini PA, Polimeni JR, Chen JE. “Mapping stimulus-driven hemodynamic changes in white matter using high-resolution fMRI at 7T”. *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [30] Taylor PA, Reynolds RC, Calhoun V, **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA, Mejia AF, Chen G. “Highlight Results, Don’t Hide Them: Improve Reproducibility (With Applications to the NARPS Results)”. *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.

- [31] Gobo V, **Gonzalez-Castillo J**, Teves JB, Bandettini PA, Kronemer SI. “Real time pupil size detection as a live marker of arousal state and perceptual sensitivity” *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [32] Kronemer SI, Holness MN, Morgan TA, Gonzalez-Castillo J, Teves JB, Handwerker DA, Bandettini PA. “The neural mechanisms of interoceptive conscious perception: A 7T fMRI study of afterimages” *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [33] Teves JB, Bandettini PA, Caballero-Gauges C, Dowdle L, DuPre E, **Gonzalez-Castillo J**, Handwerker DA, Moia S, Reddy N, Salo T, Uruñela E. “A growing multi-echo fMRI ecosystem” *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [34] Spurney M, Faskowitz J, **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA. “Edge-time series summary metrics: predictive value for demographics and personality traits” *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [35] Holness MN, Handwerker DA, Teves JB, Morgan T, Chen G, **Gonzalez-Castillo J**, Bandettini PA. “Using multi-echo fMRI to remove physiological noise during naturalistic viewing” *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [36] Gephart I, Morgan T, **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA. “Multiple factors influence perception in the audiovisual bounce effect” *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [37] Klein R, Morgan T, Huber L, Handwerker DA, **Gonzalez-Castillo J**, Bandettini PA. “Cluster-based connectivity tool: capturing laminar and columnar topography during naturalistic tasks” *29th Annual Meeting of the Organization for Human Brain Mapping*, Montreal, Canada, July **2023**.
- [38] Fan DJ, **Gonzalez-Castillo J**, Bandettini PA, Polimeni JR, Chen JE. “Mapping stimulus-driven hemodynamic changes in white matter using high-resolution fMRI at 7T”. *30th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Toronto, Canada, June, **2023**.
- [39] **Gonzalez-Castillo J**, Fernandez I, Handwerker DA, Lam KC, Pereira F, Bandettini PA. “Manifold learning and dimensionality estimation for the Human Functional Connectome”. *30th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Toronto, Canada, June, **2023**.
- [40] Shahsavarani S, Thibodeaux DN, Lodgher F, Xu W, Handwerker DA, **Gonzalez-Castillo J**, Bandettini PA, Hillman EMC. “Varying patterns of brain-wide neuronal activity underlie correlation structure of low-frequency spontaneous hemodynamic signals”. *51st Annual Meeting of the Society for Neuroscience*, San Diego, CA, USA, November **2022**.
- [41] Spurney M, **Gonzalez-Castillo J**, Lam KC, Handwerker DA, Teeves JB, Pereira F, Bandettini PA. “How conscious in-scanner thoughts modulate functional connectivity during resting-state fMRI”. *51st Annual Meeting of the Society for Neuroscience*, San Diego, CA, USA, November **2022**.
- [42] **Gonzalez-Castillo J**, Spurney M, Lam KC, Handwerker DA, Teeves J, Pereira F, Bandettini PA. “How conscious thoughts during resting-state affect functional connectivity estimates”. *28th Annual Meeting of the Organization for Human Brain Mapping*, Glasgow, Scotland, June **2022**.
- [43] Markiewicz C, DuPre E, Esteban O, Goncalves M, Gonzalez-Castillo J, Handwerker DA, et al. “Standardizing support for multi-echo fMRI data across the NiPreps ecosystem”. *28th Annual Meeting of the Organization for Human Brain Mapping*, Glasgow, Scotland, June **2022**.

- [44] Kumar S, Arzi A, Bereham C, **Gonzalez-Castillo J**, Fernandez IS, Tagliazucchi E, Mediano P, Bandettini PA, Bekinschtein T. "Sleep-related brain functional reorganization reflects task engagement". *28th Annual Meeting of the Organization for Human Brain Mapping*, Glasgow, Scotland, June **2022**.
- [45] Holness MN, Handwerker DA, Teeves JB, Moegan AT, **Gonzalez-Castillo J**, Bandettini PA. "Multi-echo fMRI denoising with physiological and motion information". *28th Annual Meeting of the Organization for Human Brain Mapping*, Glasgow, Scotland, June **2022**.
- [46] Ahmed Z, Bandettini PA, Bottenhorn K, Caballero-Gaudes C, Dowdle L, DuPre E, Gonzalez-Castillo J, et al. "tedana: multi-echo fMRI and related open tools". *28th Annual Meeting of the Organization for Human Brain Mapping*, Glasgow, Scotland, June **2022**.
- [47] Shahsavarani S, Thibodeaux D, Nwokeabia C, Xu W, Lodgher F, Handwerker D, **Gonzalez-Castillo J**, Bandettini P, Hillman. "Do arousal fluctuations alter functional connectivity?" *50th Annual Meeting of the Society for Neuroscience*, Chicago, USA, November, **2021**
- [48] Thibodeaux D, Shahsavarani S, Nwokeabia C, Xu W, Lodgher F, Handwerker D, **Gonzalez-Castillo J**, Bandettini P, Hillman E. "Investigating the neural and behavioral basis of Dynamic Functional Connectivity." *50th Annual Meeting of the Society for Neuroscience*, Chicago, USA, November, **2021**
- [49] Fernandez IS, **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA, "Monitoring vigilance during resting-state using ultra-slow fluctuations in the 4th ventricle" *50th Annual Meeting of the Society for Neuroscience*, Chicago, USA, November, **2021**
- [50] Shahsavarani S, Nwokeabia C, Xu W, Zhu H, Lodgher F, Cambareri M, Handwerker D, **Gonzalez-Castillo J**, Bandettini P, Schevon C, Hillman E. "Decoding the Neural Basis of Resting-State Functional Connectivity Mapping." *The virtual 7th Annual BRAIN Initiative Investigators Meeting*. June, **2021**
- [51] Handwerker D., Ahmed Z, Bandettini B, Bottenhorn K, Caballero-Gaudes C, Dowdle L, DuPre E, **Gonzalez-Castillo J** et al. "Tedana: multi-echo fMRI tools and resources" *27th Annual Meeting of the Organization for Human Brain Mapping*, Virtual, June **2021**
- [52] Koiso K, Handwerker DA, **Gonzalez-Castillo J**, Huber L, Glen D, Khojandi A, Chai Y, Bandettini PA, Miyawaki Y. "Neural information sources identified by sparse analysis of ultra-highspeed fMRI signals and susceptibility-weighted imaging at an ultra-high magnetic field" *44th Annual Meeting of The Japan Neuroscience Society*, Kobe, Japan, **2021**.
- [53] **Gonzalez-Castillo J**, Handwerker DA, Bandettini PA. "Amplitude of slow fluctuations in CSF as a time-resolved marker of sleep for resting-state fMRI: a validation study". *28th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Virtual, August, **2020**.
- [54] Miyawaki Y, Handwerker DA, **Gonzalez-Castillo J**, Huber L, Khojandi A, Chai Y, Bandettini PA. "Event-related decoding of visual stimulus information using short-TR BOLD fMRI at 7T" *28th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Virtual, August, **2020**.
- [55] Miyawaki Y, Handwerker DA, **Gonzalez-Castillo J**, Huber L, Khojandi A, Chai Y, Bandettini PA. "Decoding of neural information representation independent of hemodynamic delays using the ultra-fast acquisition of ultra-high field fMRI signals" *43rd Annual Meeting of the Japan Neuroscience Society*, July, **2020**.

- [56] Miyawaki Y, Handwerker DA, **Gonzalez-Castillo J**, Huber L, Khojandi A, Chai Y, Bandettini PA. "Time-resolved fast neural decoding independent variation in hemodynamic response latency" *26th Annual Meeting of the Organization for Human Brain Mapping*, Virtual, July, **2020**
- [57] Bandettini PA, Bottenhorn K, Caballero-Gaudes C, Dowdle L, Dupre E, **Gonzalez-Castillo J**, Handwerker D, Laird A, Lee J, Markello R, Moia S, Salo T, Teves J, Uruñuela E, Vaziri-Pashkam M, Whitaker K. "Tedana: multi-echo software and communal resources". *26th Annual Meeting of the Organization for Human Brain Mapping*, Virtual, July, **2020**.
- [58] DuPre E, **Gonzalez-Castillo J**, Handwerker DA, Markello R, Taylor S, Whitaker K "Tedana: robust and extensible software for multi-echo denoising" *25th Annual Meeting of the Organization for Human Brain Mapping*, Rome, **2019**.
- [59] **Gonzalez-Castillo J**, Caballero-Gaudes C, Topolski N, Pereira F, Handwerker DA, Bandettini PA "Contributions of covert self-driven cognition to resting state dynamic functional connectivity" *25th Annual Meeting of the Organization for Human Brain Mapping*, Rome, **2019**.
- [60] Handwerker DA, Sharon K, Shaik M, Thibodeaux, Montgomery MK, Zhao H, **Gonzalez-Castillo J**, Molfese PJ, Nielson D, Hillman E, Bandettini PA. "Stability of functional connectivity in mice using wide field optical imaging" *25th Annual Meeting of the Organization for Human Brain Mapping*, Rome, **2019**.
- [61] Uruñuela-Tremiño E, Moia S, **Gonzalez-Castillo J**, Caballero-Gaudes C "Deconvolution of multi-echo fMRI data with multivariate multi-echo sparse paradigm free mapping" *27th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Montreal, May, **2019**.
- [62] Gonzalez-Castillo J, Caballero-Gaudes C, Bandettini PA. "Pseudo-quantitative deconvolution of neuronal-related BOLD events with unknown timing" *48th Annual Meeting of the Society for Neuroscience*, San Diego, November, **2018**.
- [63] Xie H, **Gonzalez-Castillo J**, Handwerker D, Molfese P, Bandettini PA, Calhoun V, Mitra S. "Efficacy of different dynamic functional connectivity methods to capture cognitively relevant information". *26th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Paris, June, **2018**.
- [64] Caballero-Gaudes C, Bandettini PA, **Gonzalez-Castillo J**. "Improved detection of neuronal-related BOLD events of unknown timing with Multi-Echo Sparse Paradigm Free Mapping" *26th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Paris, June, **2018**.
- [65] Handwerker DA, **Gonzalez-Castillo J**, Nielson D, Zheng C, Molfese P, Bandettini PA. "Moving away from ICA in multi-echo fMRI denoising". *24th Annual Meeting of the Organization for Human Brain Mapping*, Singapore, June, **2018**.
- [66] Zheng C, Xie H, **Gonzalez-Castillo J**, Bandettini PA. "Robust testing of temporal dynamics in resting-state fMRI". *24th Annual Meeting of the Organization for Human Brain Mapping*, Singapore, June, **2018**.
- [67] Jangraw D, Finn ES, **Gonzalez-Castillo J**, Handwerker DA, Ghane M, Rosenberg MD, Panwar P, Bandettini PA. "Functional Connectivity-based predictor of reading recall generalized to multi-task data". *24th Annual Meeting of the Organization for Human Brain Mapping*, Singapore, June, **2018**.
- [68] **Gonzalez-Castillo J**, Topolski N, Brown J, Handwerker DA, Bandettini PA. "Spatial extent of task induced connectivity changes and its influence on whole-brain cognitive state decoding". *23rd Annual Meeting of the Organization for Human Brain Mapping*, Vancouver, Canada, June, **2017**.

- [69] Jangraw DC, **Gonzalez-Castillo J**, Handwerker DA, Panwar P, Gutierrez B, Bandettini PA. “Functional connectivity-based predictors of naturalistic reading comprehension”. *23rd Annual Meeting of the Organization for Human Brain Mapping*, Vancouver, Canada, June, **2017**.
- [70] Huber L, Handwerker DA, **Gonzalez-Castillo J**, Hall A, Jangraw DC, Guidi M, Ivanov D, Benedikt AP, Bandettini PA. “Cortical depth-dependent fMRI signal can classify sensory motor tasks”. *23rd Annual Meeting of the Organization for Human Brain Mapping*, Vancouver, Canada, June, **2017**.
- [71] Huber L, Handwerker DA, Hall A, Jangraw DC, **Gonzalez-Castillo J**, Guidi M, Ivanov D, Poser BA, Bandettini PA. “Cortical depth-dependent fMRI: heterogeneity across tasks, across participants, across days and along cortical ribbon”. *24th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Honolulu, USA. April, **2017**.
- [72] Xie H, **Gonzalez-Castillo J**, Damaraju E, Bandettini PA, Calhoun V, Mitra S. “Resting-state Functional Network Connectivity Pattern as a Cognitive Marker for Task Performance”. *24th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Honolulu, USA. April, **2017**.
- [73] **Gonzalez-Castillo J**, Panwar P, Caballero-Gaudes C, Handwerker DA, Jangraw DC, Zachariou V, Bandettini PA. “Multi-echo fMRI for rapid event related fMRI experiments”. *22st Annual Meeting of the Organization for Human Brain Mapping*, Geneva, Switzerland, June, **2016**.
- [74] Jangraw DC, Handwerker DA, **Gonzalez-Castillo J**, Panwar P, Zachariou V, Bandettini PA. “fMRI Connectivity Outperforms Magnitude in Classifying Attention to Distracting Speech While Reading”. *22st Annual Meeting of the Organization for Human Brain Mapping*, Geneva, Switzerland, June, **2016**.
- [75] Handwerker DA, Huber L, Panwar P, Gutierrez B, **Gonzalez-Castillo J**, Bandettini PA. “Cerebrovascular changes during the Valsalva Maneuver measured with VASO”. *22st Annual Meeting of the Organization for Human Brain Mapping*, Geneva, Switzerland, June, **2016**.
- [76] Huber L, Handwerker DA, **Gonzalez-Castillo J**, Jangraw DC, Ivanov D, Poser B, Goense J, Bandettini PA. “Effective connectivity measured with layer-dependent resting-state blood volume fMRI in humans”. *22st Annual Meeting of the Organization for Human Brain Mapping*, Geneva, Switzerland, June, **2016**.
- [77] Michal R, Kimmich S, **Gonzalez-Castillo J**, Popal H, White E, Martin A. “Direct modulation of aberrant social brain network connectivity in Autistic Spectrum Disorder through NeuroFeedback”. Workshop on Concepts, Actions and Objects: functional and neuronal perspectives. Roverto, Italy, May, **2016**.
- [78] Lu K, **Gonzalez-Castillo J**, Middione M, Fernandez B, Olafsson V, Kundu P, Shankaranayanan A, Liu T. “Differences in slow drift among echoes in multiband multiecho EPI data compromise TE-dependent analysis”. *24th Annual Meeting of the Society for Magnetic Resonance Imaging in Medicine*, Singapore. May, **2016**.
- [79] Huber HL, Handwerker DA, **Gonzalez-Castillo J**, Jangraw D, Marret S, Guidi N, Ivanov D, Poser BA, Goense J, Bandettini PA. Directional Connectivity measured with layer-dependent fMRI in human sensory-motor cortex. *3rd Whistler Scientific Workshop on Brain Functional Organization, Connectivity and Behavior*, Whistler-Backcomb, British Columbia, Canada, March, **2016**.
- [80] Jangraw D, Handwerker DA, **Gonzalez-Castillo J**, Panwar P, Zachariou V, Bandettini PA. BOLD functional connectivity features outperform magnitude features in classifying attention to, but

presence of, distracting speech during reading. *3rd Whistler Scientific Workshop on Brain Functional Organization, Connectivity and Behavior*, Whistler-Backcomb, British Columbia, Canada, March, **2016**.

[81] Handwerker DA, Iani G, Gutierrez B, Roopchansingh V, **Gonzalez-Castillo J**, Ungerleider LG, Bandettini PA, Pitcher D. Thetaburst TMS to the right posterior superior temporal sulcus disrupts resting state connectivity across the face-processing network as measured with multi-echo fMRI. *45th Annual Meeting of the Society for Neuroscience*, Chicago, October, **2015**.

[82] Gutierrez B, Handwerker DA, **Gonzalez-Castillo J**, Roopchansingh V, Buchanan LC, Bandettini PA. Using Multi-echo fMRI to increase contrast-to-noise and response stability. *45th Annual Meeting of the Society for Neuroscience*, Chicago, October, **2015**.

[83] **Gonzalez-Castillo J**, Buchanan LC, Handwerker DA, Roopchansingh V, Derbyshire JA, Gutierrez BE, Bandettini PA. Multi-echo Independent Component Analysis (ME-ICA) Denoising for cardiac gated fMRI. *45th Annual Meeting of the Society for Neuroscience*, Chicago, October, **2015**.

[84] Jangraw DC, Handwerker DA, **Gonzalez-Castillo J**, Gutierrez B, Roopchansingh V, Bandettini PA. Multi-echo fMRI enhances reliability of brain-wide BOLD responses to a naturalistic movie. *45th Annual Meeting of the Society for Neuroscience*, Chicago, October, **2015**.

[85] Gutierrez B, Handwerker DA, **Gonzalez-Castillo J**, Roopchansingh V, Buchanan LC, Bandettini PA. Effects of multi-echo based Denoising on reliability of a massively repeated block design task. *21st Annual Meeting of the Organization for Human Brain Mapping*, Hawaii, June, **2015**.

[86] **Gonzalez-Castillo J**, Buchanan LC, Handwerker DA, Roopchansingh V, Derbyshire J, Bandettini PA. ME-ICA Denoising of cardiac gated data (Block design). *21st Annual Meeting of the Organization for Human Brain Mapping*, Hawaii, June, **2015**.

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